

Saber Gholami

🔧 Sr. Software Engineer

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👤 Professional Profile

I am an accomplished Software Engineer with a robust technical and theoretical foundation. My expertise in algorithms and data structures for Place and Route within FPGA CAD tools and optimization passes for AI compilers such as MLIR and TVM is the result of substantial experience in professional environments. Holding a Ph.D. in Computer Science, I have adeptly tackled a wide range of complex problems, consistently delivering optimized solutions or fast heuristics.

🏢 Work Experience

Tenstorrent AI ULC

May 2025 - Present: *Sr. Engineer, AI Compilers*

- Contributed to the design and implementation of AI compilers for Tenstorrent hardware, such as blackhole and wormhole.
- Pushed the generality of the compiler and added support in the optimizer for more operators and graphs. This automation resulted in significant reduction in developers' time.
- Worked on ttnn-jit frontend and made it pip install-able for easier usage.
- Developed graph-based approaches for generating ttnn IR from a decorated python function to be used in ttnn-jit. This added support for TMs and reduction ops, as well as dynamically decomposing composite ops.

Huawei Technologies Canada Co. Ltd

Sep 2024 - May 2025: *Software Engineer*

- Contributed to the design and implementation of AI compilers tailored for Huawei's hardware platforms.
- Developed an end-to-end solution integrating MLIR with TVM, leveraging MLIR's optimization passes specifically optimized for Huawei hardware, alongside TVM's auto-scheduling.
- Designed new MLIR dialects compatible with MLIR's Tosa and TVM's Relax dialects, and implemented lowering passes for these dialects.
- Led a team of engineers to design and implement an end-to-end testing infrastructure, enabling validation of multiple flows with automated C code inference capabilities using C++, Python, Shell scr.

Huawei Technologies Canada Co. Ltd

Apr 2023 - Sep 2024: *Software Engineer*

- Engaged in the development of FPGA CAD tools using C++ for Place and Route team.
- Conducted thorough research to identify cutting-edge EDA algorithms tailored to support custom architectures, and developed multiple algorithms focusing on core routing, parallel scheduling and filter generation, leading to a notable 12x acceleration in router flow through the implementation of parallel techniques.
- Designed the complete clock solution for the tool, achieving a significant 16x speed enhancement compared to the previous solution for placing and routing on the clock network.
- Collaborated closely with the Timing team to pioneer advancements in clock skew optimization and the resolution of hold violations, resulting in a noteworthy 5% increase in fmax.

Virtual Reality Lab, K.N.Toosi University, Iran Sep. 2015 - Aug. 2016:

Software Engineer

- Developed a Java framework for optimizing the movement of virtual cars in curved highways and bridges using Bezier curve fitting methods.

🎓 Education

Ph.D in Computer Science

Sep. 2019 – Dec. 2022: *Concordia University*

- Developed a novel Genetic Algorithm framework for fast message dissemination in networks with limited memory in Python.
- Designed the optimal algorithm for broadcasting in various communication networks.
- Developed a fast algorithm for community detection in social networks based on centrality measures.

- Suggested a memory-efficient model for communication in graphs.

M.Sc in Computer Engineering

Sep. 2017 – Sep. 2019: *Amirkabir University*

- Developed a novel hybrid learning model based on learning automata with applications in the dropout phase of Neural Networks in Python.
- Developed a fast algorithm for influence maximization in social networks based on graph coloring in Python.
- Developed various ML and NLP algorithms for classifying Google Play applications in Python.

⚙️ Technical Skills

- 🔗 Programming Languages: C++, C, Java, Python
- 🔗 Production Tools: Git, CMake, Ninja, Agile, bash, gcc, gdb, gcov.
- 🔗 AI compilers: MLIR, TVM.
- 🔗 Operating Systems: Windows, Linux
- 🔗 ML and DL: Scikit-learn, NumPy, SciPy, Pandas, Spektral, NetworkX, Nltk
- 🔗 Web technologies: HTML, CSS, Bootstrap, Javascript, jQuery, Django

🌟 Recent Publications

- 📖 A Note to Non-adaptive Broadcasting, 2024
- 📖 Optimal Broadcasting in Fully Connected Trees, 2023
- 📖 HLA: a novel hybrid model based on fixed structure and variable structure learning automata, 2023
- 📖 HUB-GA: A heuristic for universal lists broadcasting using genetic algorithm, 2023
- 📖 Broadcast graphs with nodes of limited memory, 2023
- 📖 Fully-adaptive model for broadcasting with universal lists, 2022
- ❓ A complete list available at my [Goole Scholar] page.

📖 Teaching Experience

- 🖥 @ **John Abbott College, Montreal, Canada:** Foundations of Web Development
- 🖥 @ **Concordia University, Montreal, Canada:** Data Structure and Algorithms - Object-Oriented Programming - Intro to Theoretical Computer Science - Formal Methods for Software Engineering - Software Engineering - Software Design Methodologies
- 🖥 @ **Amirkabir University of Technology, Tehran, Iran:** Algorithm design - Data structure
- 🖥 @ **K.N.Toosi University of Technology, Tehran, Iran:** Algorithm design - Automata theory, languages, and computation - Logic circuit

👍 References

My current supervisor @ *Tenstorrent AI ULC*

- 👤 Xander Chin
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My previous supervisor @ *Huawei Technologies Canada Co. Ltd.*

- 👤 Mark Bourgeault
- 🌐 LinkedIn/mark-bourgeault
- ✉ mark.bourgeault@huawei.com